The Graded Net: Klein-Charged Currents, the Jordan Diagonality of Observables, and the Selection Rule as Scattering

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Abstract. The generation-grading paper proved the Klein selection rule at the algebra level; this note represents it on the net, where the multiplicity structure and the interaction machinery meet in one object. The graded net consists of fifteen mutually commuting chiral currents j^A, one per imaginary sedenion direction, carrying the Klein charge gen(A) ; the free theory is the series' construction componentwise. The vertex tensor $g_{ABC} = \langle e_A e_B, e_C \rangle$, computed from the independently rebuilt multiplication table, has exactly the 210 Klein-conserving triples as its support, is antisymmetric in (A, B), and is invariant under the exhibited generation automorphism τ — all verified entry by entry. The antisymmetry carries a theorem: since distinct imaginary units anticommute, the symmetrized (Jordan) products vanish identically off the diagonal (verified: maximum 0), so **the measurement channel of this series is generation-diagonal — no observable changes generation number, and generation transitions are exclusively dynamical**, carried by the antisymmetric channel: the measurement-split and generation-grading papers meeting in one line. Dynamically, transitions are realized by background-mediated derivative couplings $H = \lambda \int \mu \, M_{AB} \, j^A \partial j^B$ with $M = B^C g_{C..}$ — self-adjoint in one line, hence symplectic — and the scattering verifies the algebra's predictions exactly: the τ-cycle transition amplitudes are *bit-identical* (0.111316 for $1 \to 2$ via a grade-3 background, $2 \to 3$ via grade-1, $3 \to 1$ via grade-2), Klein-forbidden channels vanish identically ($1 \to 3$ via grade-3: 0.0), a background of Klein charge k mixes only pairs with $A \oplus B = k$ (core backgrounds are generation-preserving), and memory windows are inert at the 10^{-9} filter floor — the measurement gate's theorem holding componentwise: *generation transitions never pump memory*. The model-building rulebook is thereby assembled: Klein-charge conservation, Jordan diagonality of observables, τ-invariance, gauge universality, and the sector taxonomy — every rule a verified theorem of the archive.

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1. The step

The charged-coupling paper closed the sector taxonomy and handed the pattern question to the multiplicity structure; the generation-grading paper answered with three theorems at the algebra level. What remained was representation: the graded structure realized on the net itself, with the selection rules appearing where physics lives — in scattering.

2. The graded net

For each imaginary direction $A \in \{1, \ldots, 15\}$ of $\mathbb{S}$, let j^A be a copy of the series' chiral current; distinct components commute. The Klein charge of the component is gen$(A) = A \bmod 4$; the four grade sectors are the core currents ($A \in \{4, 8, 12\}$) and the three generation families. The free theory — state, type III_1 structure, local net, cone assembly — is the corpus componentwise, and needs no re-derivation.

3. The vertex tensor, and the Jordan diagonality of observables

The algebra supplies the couplings: $g_{ABC} = \langle e_A e_B, e_C \rangle$, computed from the multiplication table rebuilt from the doubling recursion. Verified properties: the support is *exactly* the 210 Klein-conserving triples $\{(A, B, A \oplus B) : A \neq B\}$; g is antisymmetric in (A, B) ($\max|g + g^T| = 0$); and every coupling is invariant under the generation automorphism τ of the companion paper.

The antisymmetry is not bookkeeping — it is a physical theorem:

Theorem (Jordan diagonality). Distinct imaginary sedenion units anticommute (verified: $\max|e_A e_B + e_B e_A| = 0$ for $A \neq B$), so the Jordan products $e_A \circ e_B$ vanish off the diagonal. The symmetric channel — identified across this series as the *measurement* channel — is therefore generation-diagonal: **no observable changes generation number.** Generation transitions live exclusively in the antisymmetric, dynamical channel. ■

Measurement preserves flavor; only dynamics changes it — a statement the algebra enforces, joining the measurement-split paper's kinematics to the generation grading in a single line.

4. The selection rule as scattering

Transitions are realized by coupling the graded currents to a classical background B^C through the algebra's own vertex:

$$H = \lambda \int \mu(u)\, M_{AB}\, j^A\, \partial j^B\, du, \qquad M_{AB} = B^C g_{CAB},$$

the derivative required precisely because the plain bilinear is symmetric and the algebra's transition content is antisymmetric. H is self-adjoint in one line (components commute; the antisymmetric contraction of the symmetrized product is a total derivative), so the induced flow on smearings is symplectic; it was integrated with band-filtered spectral derivatives, and the verified scattering reads the algebra back exactly:

- **The τ-cycle is exact at the S-matrix level.** The transition amplitudes $|1 \to 2|$ via a grade-3 background, $|2 \to 3|$ via grade-1, and $|3 \to 1|$ via grade-2 are *bit-identical*: $0.111316, 0.111316, 0.111316$. The $\mathbb{Z}_3$ generation symmetry, exhibited algebraically in the companion paper, holds in the dynamics to every computed digit.
- **Klein-forbidden channels vanish identically.** $|1 \to 3|$ via a grade-3 background: 0.0 — not small, zero: the background of charge k mixes only pairs with $A \oplus B = k$, so core backgrounds are generation-preserving and each charged background opens exactly its own transition pairs.
- **Generation transitions never pump memory.** The plateau memory window leaks 8.6×10^{-9} (the spectral filter's floor) under the full internal mixing: the measurement gate's sector theorem holds componentwise, so flavor-changing scattering conserves every memory label.

Numerical honesty: the discrete symplectic form is conserved to 1.4% over the evolution, a residual of the band filter's non-commutativity with the coupling profile; the analytic symplecticity is exact by the self-adjointness above, and the three structural results — which are the theorems — are exact as computed.

5. The rulebook assembled

Charged sources per component pump the Im $\mathbb{S}$-valued memory $m^A = -2\pi\lambda Q^A$, composing additively, with the Klein charge conserved through every algebra-mediated process. The model-building rulebook of the interaction era now stands complete, every rule a verified theorem: **Klein-charge conservation** (the grading), **Jordan diagonality** (observables preserve generation), τ**-invariance** (the cyclic symmetry, exact in scattering), **gauge universality** (the democratic octonion), and the **sector taxonomy** (bilinear and current-type couplings preserve, charged sources pump $-2\pi\lambda Q$). What the algebra fixes, it fixes exactly; what it leaves free — coupling values, hierarchies, the choice of backgrounds — it hands to the state, as this series has found at every level.

6. Scope

Five items. *(i)* Backgrounds are classical here; promoting them to quantized mediator fields is genuine interacting QFT and the era's continuing construction program. *(ii)* No identification with Standard-Model flavor is claimed; the adjacency remains with the companion papers on the three-generation construction. *(iii)* The transition amplitudes' common *value* is background- and profile-dependent; only their equalities and zeros are algebra-fixed. *(iv)* The discrete-symplecticity residual is a filter artifact, documented; a manifestly symplectic discretization is a numerical refinement, not a physics gap. *(v)* Hierarchy and mixing values remain consigned to the state stratum.

References

- Companion papers: *The Generation Grading: The Klein Selection Rule, Democratic Gauge Universality, and the Cyclic Generation Symmetry*; *The Two Halves of One Product: The Jordan/Malcev Split of the Octonions and the Measurement Channel*; *The Measurement Gate: The Probe Scheme on the Net, Unsharp Induced Observables, and Sector-Preserving Scattering*; *The Charged Coupling: The Infrared Catastrophe as a Power Law, Sector Pumping, and the Memory–Charge Relation.*

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.